

Stationary Balance Instrument

Freeze one estimate; keep live evidence moving

E0 SYNTHETIC REPLAY — NO POWERED INERTIAL MODULE CLAIM

Lesson	045 — project	Time	120–180 minutes
Prerequisites	Lessons 043–044	Board	compile-only Mega replay
Evidence	named memory result cells	Boundary	no bus, pins, wiring, or endpoints

Project boundary. This is a stationary, hand-tilted tabletop instrument built from copied synthetic values and pure presentation intent. It is not a balancing controller, handheld tracker, navigation aid, vehicle instrument, safety device, or powered sensor demonstration.

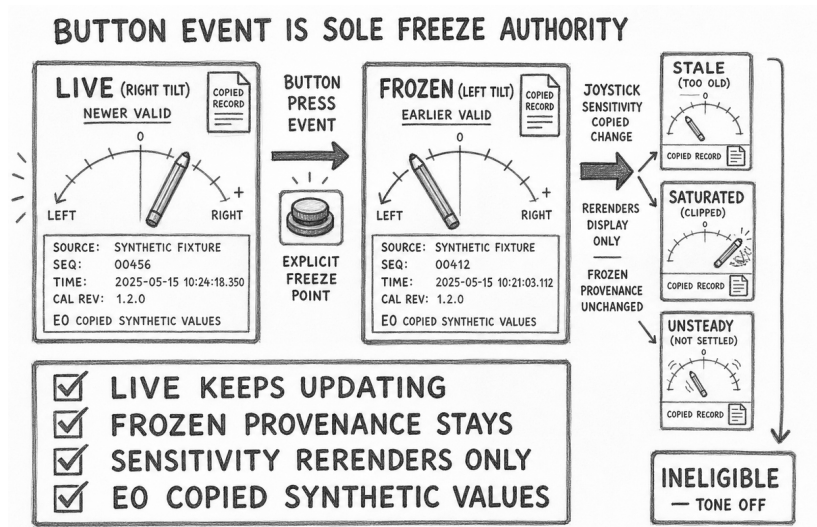
1 Compose one atomic frame

One `BalanceInstrumentInput` carries a copied inertial observation, joystick observation, freeze-button observation, frame time, and frame sequence. The caller snapshots each producer once. The instrument validates the complete frame before committing any state.

Copied producer	Identity	Required checks
inertial	source, time, sequence, configuration and calibration	freshness contract, readiness, quality, saturation, status
joystick	time and sequence	axes in $-1000 \dots 1000$ permille, event agrees with signs, status
button	time and sequence	pressed/event tuple is self-consistent, status
frame	frame time and sequence	forward modular order, no future producer, bounded input skew

An exact delta-zero replay is idempotent. A changed field at the same identity is malformed. A later frame may reuse the same immutable inertial sample, but the instrument recomputes its age, freshness quality, sequence gap, and derived status at the new frame time. It never fabricates a new sample.

2 Follow live and frozen evidence

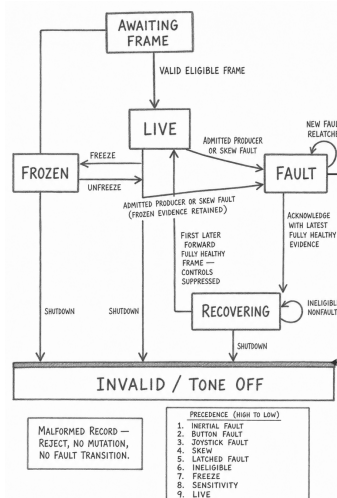


Alternative text: a pencil comparison shows current copied inertial evidence feeding a live orientation while a button press copies the current eligible estimate into a separate frozen evidence card. Later live tilt and age continue changing beside the retained frozen comparison.

Only a valid **pressEvent** is freeze authority. Tilt, sensitivity input, a held button, and a replayed press cannot freeze. A valid press uses the current frame's prepared estimate, not the previous snapshot. A second valid press unfreezes. An invalid current estimate can never become frozen.

Step	Predict	Observe in named result cells
level frame	Live and eligible level estimate	live evidence available; frozen unavailable
press with right tilt	Frozen	frozen evidence equals this frame's right estimate
later forward tilt	frozen comparison remains right	live evidence changes to forward
sensitivity increase	frozen estimate identity stays fixed	frozen presentation is re-rendered at new sensitivity
unfreeze press	Live	current live evidence selects presentation

3 Trace the state machine



Alternative text: a pencil state flow starts at AwaitingFrame, moves between Live and Frozen only on eligible frames and explicit button presses, latches Fault on an admitted producer or skew fault from either mode, and requires acknowledgement plus a later healthy forward frame before returning through Recovering. Malformed records reject without entering Fault.

Mode	Meaning
AwaitingFrame	initialized, but no complete frame is accepted
Live	current eligible estimate selects presentation intent
Frozen	retained eligible estimate selects presentation while live evidence still updates
Fault	a classified failure is latched; fault intent is tone-off
Recovering	acknowledgement succeeded; a later healthy forward frame is still required

A healthy frame does not silently clear **Fault**. **acknowledgeFault()** preserves evidence, sensitivity, replay identity, and diagnostic epoch. The exact pre-ack replay cannot complete recovery. A forward healthy frame does, while suppressing simultaneous freeze and sensitivity events. Shutdown clears retained evidence and publishes the configured invalid, tone-off intent.

4 Adjust sensitivity without changing evidence

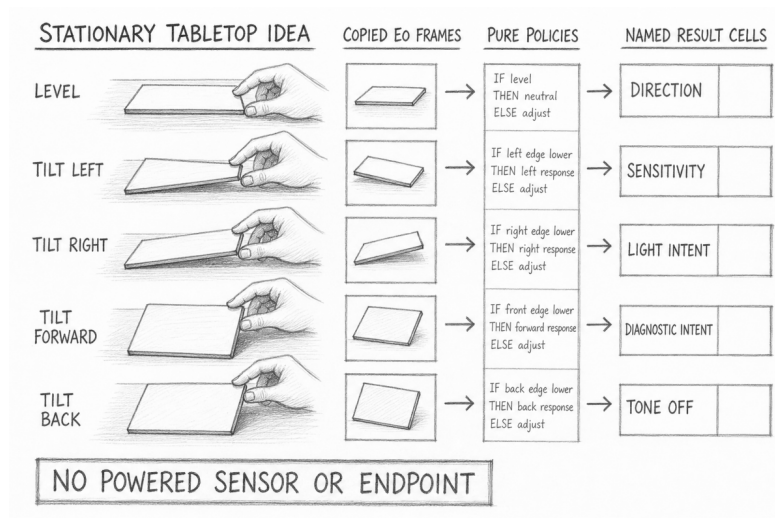
The joystick event changes only presentation sensitivity. Positive-only axes mean **Increase**; negative-only axes mean **Decrease**; zero axes mean **None**. Mixed positive and negative signs are an admitted **Contradictory** no-op and make the complete frame not fully healthy.

$$s_{\text{next}} = \text{clamp}(s \pm s_{\text{step}}, s_{\text{min}}, s_{\text{max}}) \text{ permille.}$$

current (permille)	event	step (permille)	next (permille)
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Changing sensitivity never changes pitch, roll, source identity, timestamp, sequence, or frozen provenance. It asks the Lesson 044 presentation policy to render the selected estimate again. Every tone remains copied intent; this E0 lesson drives no sounder.

5 Run the staged synthetic experiment



Alternative text: a pencil experiment strip stages startup, level, cardinal tilts, sensitivity clamps, freeze, changed live tilt, unfreeze, stale input, producer failure, acknowledgement, healthy recovery, saturation, timestamp rollover, and shutdown as copied memory records.

1. Initialize. Predict **AwaitingFrame** with the configured tone-off presentation.
2. Replay level and four cardinal tilts. Predict one complete live orientation and presentation record per accepted frame.
3. Increase and decrease sensitivity through both clamps. Predict no wraparound and no change to measurement evidence.
4. Freeze an eligible estimate, change the live tilt, and unfreeze. Predict distinct live and frozen evidence during comparison.
5. Reuse an inertial sample in a later frame beyond its maximum age. Predict **Stale**, unavailable presentation eligibility, and tone-off intent without a fabricated producer event.
6. Inject producer failure and acknowledge it. Predict **Fault**, then **Recovering**; require a later healthy forward frame for **Live**.

7. Replay saturation, timestamp rollover, exact duplicates, and shutdown. Predict stable identity, bounded intent, and no partial mutation.

6 Diagnose before interpreting

Observation	Likely classification	Interpretation
not ready or too old	Stale	accepted payload may remain, but it is ineligible now
canonical axis limit reached	Saturated	do not interpret an angle or enable tone
gravity/rate guard fails	Unsteady	stationary-orientation assumption failed
source/configuration mismatch	invalid configuration	freshness or provenance contract disagrees
future time or changed replay identity	malformed frame	reject without mutating accepted state
producer age beyond input skew	Fault	commit canonical safe-state evidence and advance replay identity
producer non-OK status	Fault	retain compact evidence and publish fault-safe tone-off intent

Evidence worksheet

frame time (ms)	sample time (ms)	age (ms)	quality/status
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
frame sequence	sample sequence	live direction	frozen direction
_____	_____	_____	_____
_____	_____	_____	_____

7 Bound the E0 software resources

AVR GCC 7.3 measurements use fixed storage and no heap, RTTI, exceptions, virtual tables, Wire, or TWI. The application owns no timer or interrupt; the linked Arduino startup/runtime still includes its ordinary timer-0 support.

Measured item	bytes	Measured item	bytes
BalanceInstrumentConfig	76	BalanceInstrumentInput	101
caller replay storage	102	BalanceInstrumentOutput	119
inertial observation policy	80	orientation policy	38
presentation policy	95	BalanceInstrument	339
resident composition	521	worst live composition	741

The 339-byte instrument is below its revised 352-byte target and 384-byte hard limit. Resident and worst-live results are below their 560/688-byte and 816/976-byte target/hard gates. The final expanded canonical sketch measures 26,398 bytes of flash (budget 28,672) and 1,898 bytes of static SRAM (budget 2,048).

The final no-LTO link measures 21,538 bytes of `.text`, 238 bytes of `.data`, and 1,660 bytes of `.bss`; the

latter two total the 1,898-byte static use. Compiler call-graph evidence bounds the foreground at 851 bytes. Conservatively adding the linked timer-0 interrupt's 15 bytes gives 866 bytes, leaving 5,428 bytes of SRAM—4,404 bytes above the 1,024-byte margin gate. This is compiler and link evidence, not a runtime stack canary or physical measurement. Runtime-canary evidence remains deferred to E1.

8 Separate acquisition from safe state

Acquisition prediction. A valid synthetic frame has recognized source and status values, nonzero identities and freshness revision, bounded producer skew, and self-consistent copied controls.

Acquisition observation. Inspect `acceptedFrameSequence`, `acceptedFrameAt`, producer statuses, compact live provenance, quality, age, and availability in the host harness. Do not use Serial, a powered board, or a hardware debugger.

Acquisition interpretation. Advancement means the complete copied frame passed validation. It does not prove an inertial module, joystick, button, clock, bus, or physical endpoint was acquired.

Safe-state prediction. Fault, ineligible quality, and shutdown each publish a complete configured light record with tone disabled. Shutdown also clears replay storage and retained measurement availability.

Safe-state observation. Inspect the copied mode, presentation, tone-enabled field, live/frozen availability, and replay-storage availability after each transition.

Safe-state interpretation. A canonical tone-off memory record proves the pure policy result only. No physical LED, sounder, sensor pin, or supply was made safe by this E0 replay.

9 Open physical acceptance card

Leave every field blank. No powered inertial module, joystick, button, RGB LED, diagnostic LED, sounder, bus, pin, resistor, current, timing cadence, stack peak, or electrical shutdown has been observed for this lesson. Compilation and synthetic replay are not bench acceptance.

Exact specimen and carrier markings	_____
Primary datasheet/register sources	_____
Supply and logic-level qualification	_____
Powered schematic review record	_____
Measured rail/current and instruments	_____
Physical observation and interpretation	_____
Independent power-removal safe state	_____
Reviewer, date, and acceptance record	_____

A future E1 path requires exact, separately qualified hardware and an electrically authoritative formal schematic. MPU-6050 qualification does not qualify QMI8658, or vice versa. This E0 lesson publishes no wiring table, address, register transaction, pin assignment, test point, or powered support claim.

Any future platform must be lightweight, stable, entirely supported by the table, and free of loose loads, sharp edges, pinch points, actuators, and stored energy. The project never controls a launcher, vehicle, balancing load, or safety mechanism.

10 Software evidence and further work

Deterministic tests cover lifecycle, atomic validation, semantic replay, freeze authority, sensitivity clamps, freshness, precedence, recovery, rollover, permutations, storage lifetime, and shutdown. The canonical Mega sketch contains only copied fixture and result cells; compilation is packaging evidence and the E0 sketch must not run on a powered board.

- Canonical compile-only Lesson 045 sketch.
- Deterministic balance-instrument tests.
- Balance-instrument interface.
- Lessons 043–045 design record.
- Deterministic testing contract.
- PDF visual policy.
- Safety model.

This untagged print edition has a searchable HTML alternative. It makes no PDF/UA or physical accessibility claim.